consumer etc.

For example, comfort could be expressed in abstract units or in currency. A water heater that is heated earlier in the day is likely to lose some of its heat before it is needed, which could require a top-up heating event to occur later in the day (which may incur additional cost).

If the ESA cannot calculate its cost for any reason (such as losing its connection to a Price server) it may omit this field. This is treated as extra meta data that an EMS may use to optimize a system.

9.2.7.14.15. MinPowerAdjustment Field

This field SHALL indicate the minimum power that the appliance can be requested to use.

For example, some EVSEs cannot be switched on to charge below 6A which may equate to ~1.3kW in EU markets. If the slot indicates a NominalPower of 0W (indicating it is expecting to be off), this allows an ESA to indicate it could be switched on to charge, but this would be the minimum power limit it can be set to.

9.2.7.14.16. MaxPowerAdjustment Field

This field SHALL indicate the maximum power that the appliance can be requested to use.

For example, an EVSE may be limited by its electrical supply to 32A which would be ~7.6kW in EU markets. If the slot indicates a NominalPower of 0W (indicating it is expecting to be off), this allows an ESA to indicate it could be switched on to charge, but this would be the maximum power limit it can be set to.

9.2.7.14.17. MinDurationAdjustment Field

This field SHALL indicate the minimum time, in seconds, that the slot can be requested to shortened to.

For example, if the slot indicates a NominalPower of 0W (indicating it is expecting to be off), this would allow an ESA to specify the minimum time it could be switched on for. This is to help protect the appliance from being damaged by short cycling times.

For example, a heat pump compressor may have a minimum cycle time of order a few minutes.

9.2.7.14.18. MaxDurationAdjustment Field

This field SHALL indicate the maximum time, in seconds, that the slot can be requested to extended to.

For example, if the slot indicates a NominalPower of 0W (indicating it is expecting to be off), this allows an ESA to specify the maximum time it could be switched on for. This may allow a battery or water heater to indicate the maximum duration that it can charge for before becoming full. In the case of a battery inverter which can be discharged, it may equally indicate the maximum time the battery could be discharged for (at the MaxPowerAdjustment power level).